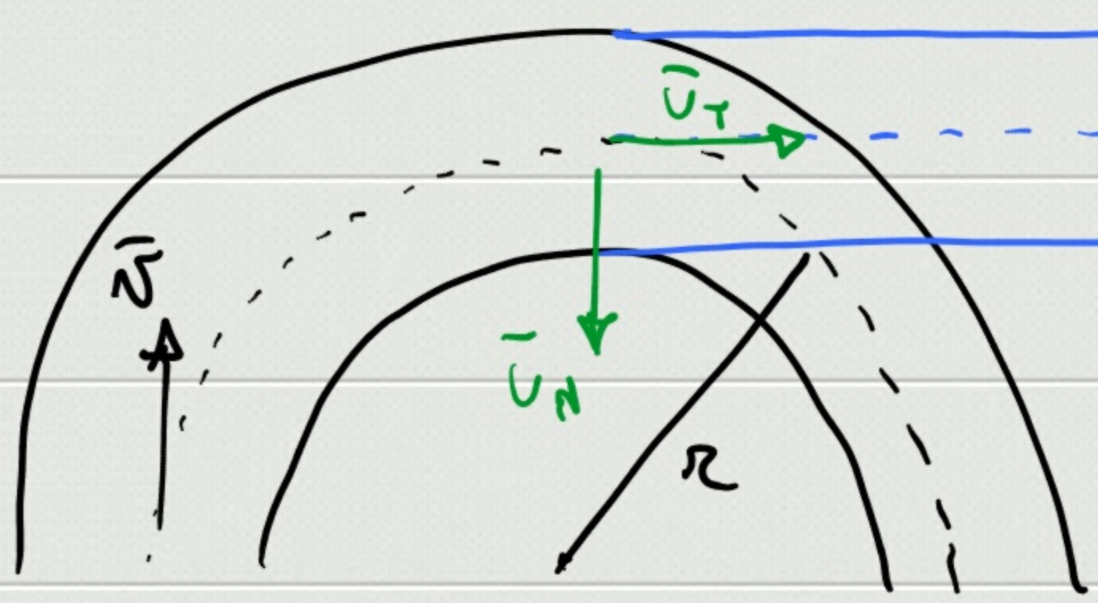
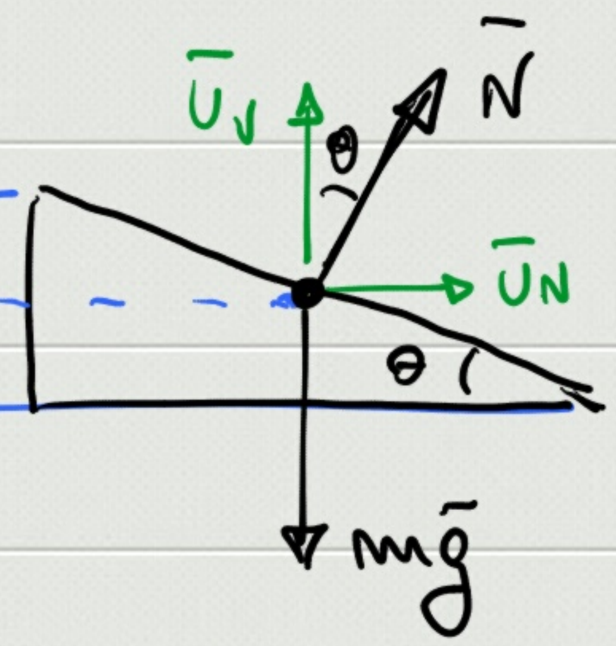
$$\bar{F} = m\bar{a} = m\bar{a}_T + m\bar{a}_N = \underbrace{m\frac{dv}{dt}}_{F_T}\bar{u}_T + \underbrace{m\frac{v^2}{R}}_{F_N}\bar{u}_N$$





$v = |\bar{v}| = \text{cost}$

liscio

moto circolare uniforme : $a_T = 0$ $\quad a_N \neq 0$

$$m\bar{g} + \bar{N} = m\bar{a} = m\bar{a}_N$$

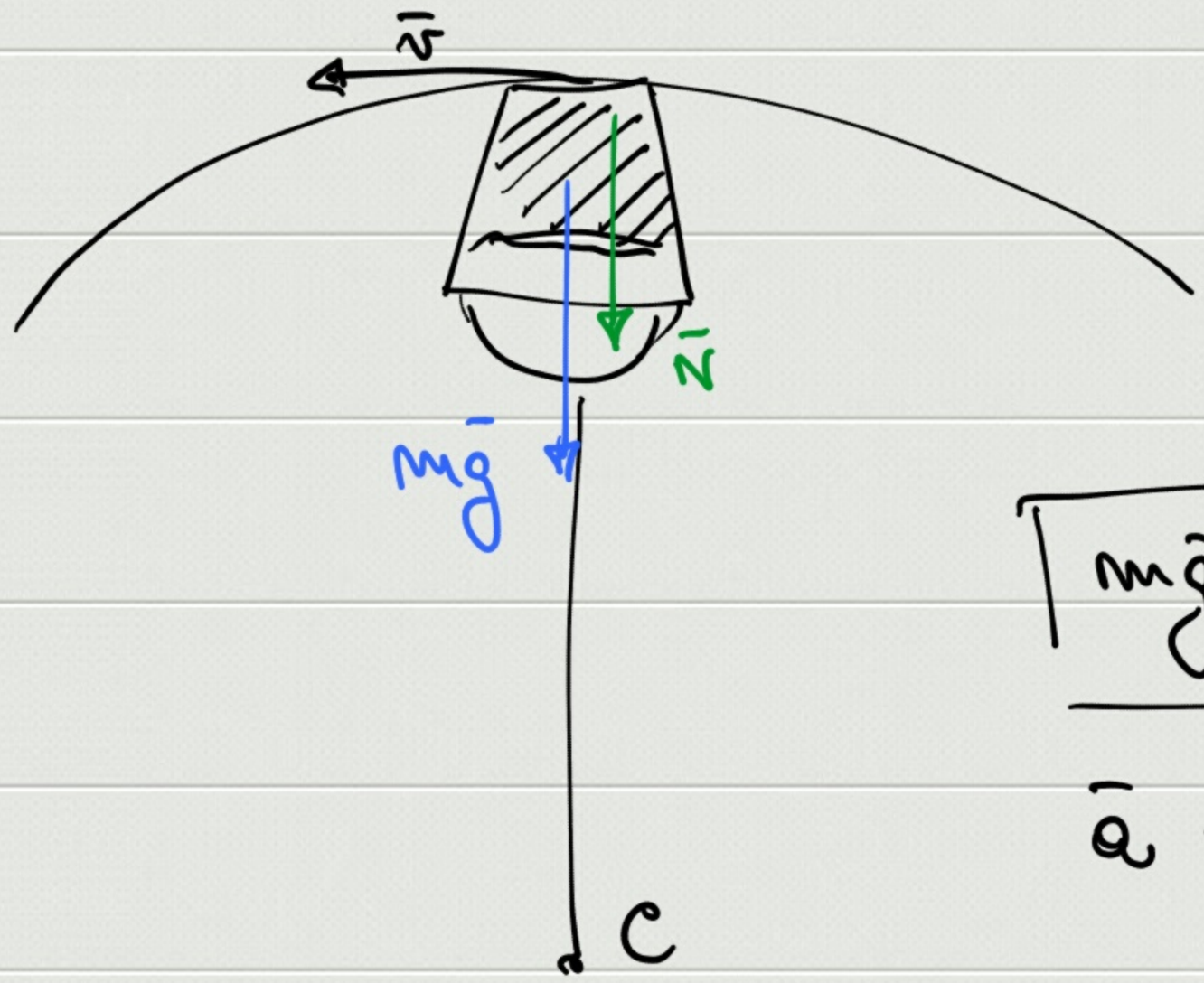
$\bar{u}_N$: $N \sin\theta = m\dfrac{v^2}{r}$ $\Rightarrow$

$\bar{u}_V$: $N\cos\theta - mg = 0 \Rightarrow N = \dfrac{mg}{\cos\theta}$

$$\Rightarrow \cancel{m}g\,\mathrm{tg}\,\theta = \cancel{m}\frac{v^2}{r} \Rightarrow \boxed{\mathrm{tg}\,\theta = \frac{v^2}{rg}}$$

$v = 120\ \text{km/h} = 33.3\ \text{m/s}$ $\quad r \sim 20\ \text{m}$

$\Rightarrow \theta = 80°$

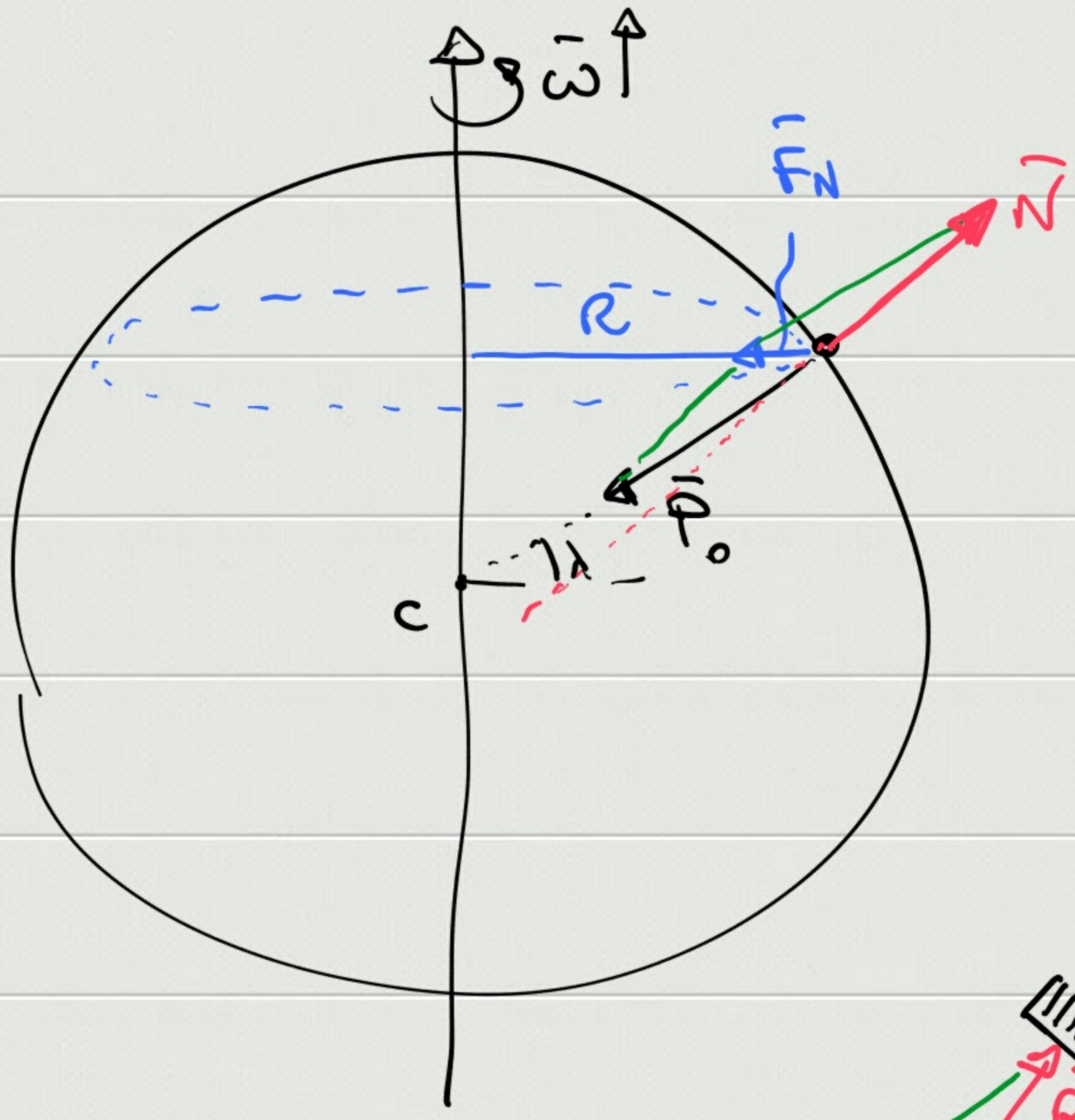


$$\boxed{m\bar{g} + \bar{N} = m\bar{a}}$$

$$\bar{a} = \bar{a}_N$$

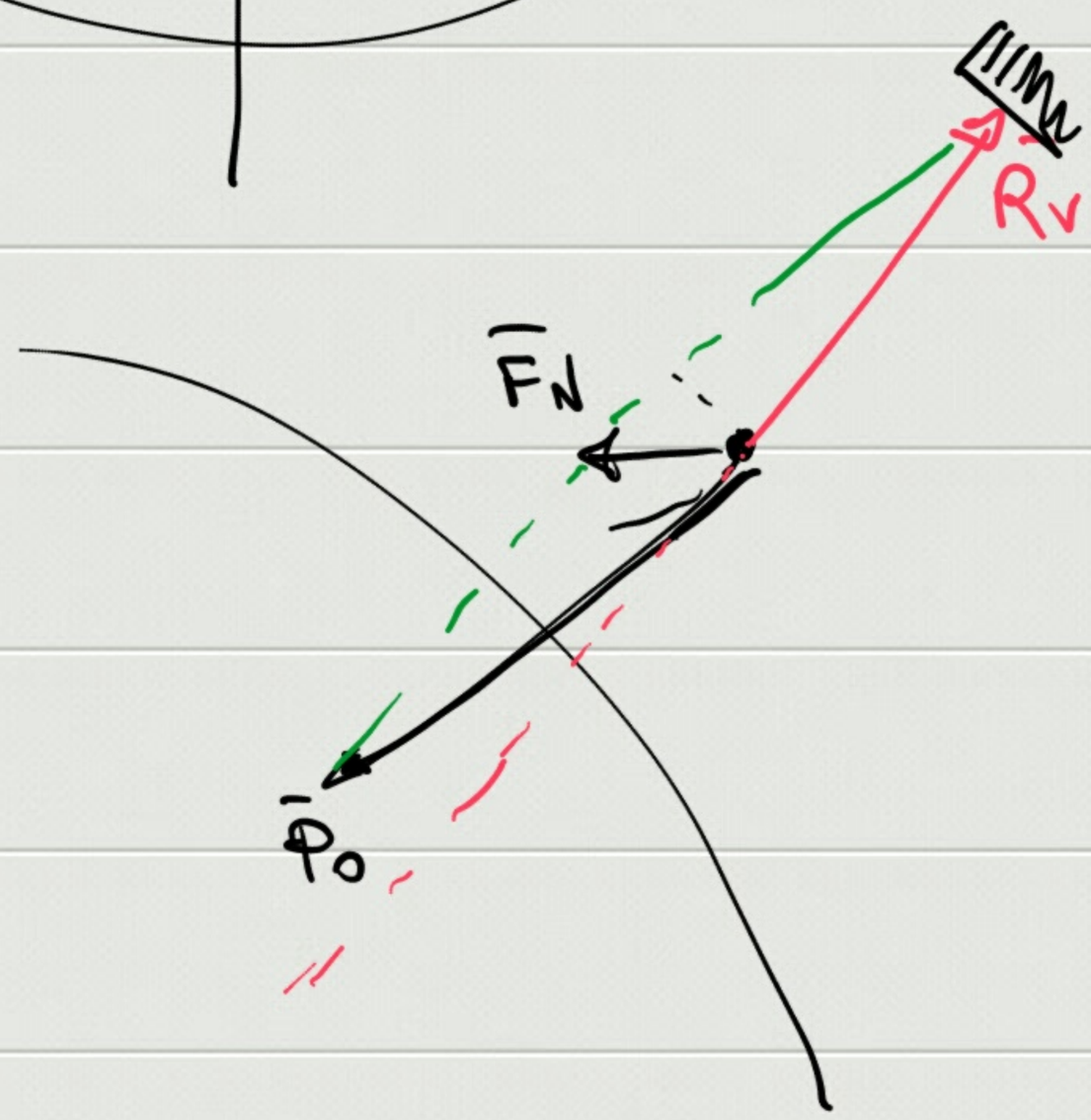
$$mg + N = m\frac{v^2}{R} \Rightarrow N = m\left(\frac{v^2}{R} - g\right) > 0$$

$$v^2 > gR \Rightarrow v > \sqrt{gR} = v_{min}$$



$$\bar{P}_0 + \bar{R}_V = \underbrace{m\,\bar{a}_N}_{\bar{F}_N}$$

$$\boxed{\bar{P} = -\bar{N}}$$



$$a_N = \omega^2 R = \frac{2\pi}{24h} R_T \cos\lambda \simeq 0.034 \cos\lambda \ m/s^2$$